POKÉMON: SHINY VICE

SYSTEM DESIGN SPECIFICATION (v4.0)

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Deployment Context: Core Architecture Engine & ROM Hacking Specification
Base Framework: Source Decompilation Engine (pokeemerald-expansion)
Target System: Game Boy Advance Hardware / Emulator (mGBA Core Specification)
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1. Executive Summary & Core Pillars

Pokémon: Shiny Vice is a structural modification of the Generation III Game Boy Advance architecture, reimagined from the ground up as a dedicated competitive ecosystem for the 'Shiny Hunting' (Shunting) community. Unlike generic ROM hacks that rely on randomizers or trivial cheat codes, Shiny Vice treats shiny collection as a strategic, merit-driven progression system.

The design revolves around a strict three-phase macroscopic gameplay contract:

- **Defensive Preparation:** Players must breed, catch, and configure highly dedicated utility catchers equipped with custom tracking equipment.
- **Brutal Competitive Combat:** Gating critical progression items behind intense 6v6 boss parameters modeled after high-tier difficulty overhauls.
- **Targeted Odds Optimization:** Utilizing earned economic resources to buy custom multipliers down to a linear global hard cap.

2. Base Engine & Compiler Dependency Specifications

The project completely bypasses archaic binary injection tooling in favor of a full modern Source Code Decompilation project, leveraging the human-readable reverse-engineered C codebase of Pokémon Emerald. This methodology removes hardcoded layout constraints and memory limitations.

To establish the compiler pipeline natively within a Linux environment (such as Nobara Linux), run the following configuration sequence:

```
sudo apt install build-essential binutils-arm-none-eabi \
    gcc-arm-none-eabi libnewlib-arm-none-eabi git libpng-dev -y
git clone https://github.com/rh-hideout/pokeemerald-expansion.git
cd pokeemerald-expansion
make -j$(nproc)
```

3. The Additive Roll-Budget Contract & Save Block Allocations

To protect the value and psychological dopamine loop of hunting, shiny occurrences do not rely on raw multiplier spikes, which trigger cascading calculation errors. Instead, the game adopts a linear **Roll-Budget Model**. Extra attempts generated by charms or chains are cleanly fed into a single function capped at a strict threshold of **8 total attempts (1/512 absolute odds)**.

The custom implementation below replaces the core verification engine inside src/pokemon.c:

```
bool8 IsPokemonShiny(struct Pokemon *mon)
{
    u32 otId = GetMonData(mon, MON_DATA_OT_ID, NULL);
    u32 personality = GetMonData(mon, MON_DATA_PERSONALITY, NULL);
    u16 trainerId = otId & 0xFFFF;
    u16 secretId = otId >> 16;
    u8 totalRolls = 1; // Base unboosted roll

    if (CheckBagHasItem(ITEM_SHINY_CHARM_TIER_1, 1)) totalRolls += 1;
    if (GetActiveChainBonus() > 0) totalRolls += GetActiveChainBonus();

    if (totalRolls > 8) totalRolls = 8; // Enforce hard contract ceiling

    for (u8 i = 0; i < totalRolls; i++)
    {
        u32 shinyValue = (trainerId ^ secretId) ^ (personality ^ (personality >> 16));
        if ((shinyValue & 0xFFF0) == 0) return TRUE;
        personality = Random32(); // Generate pseudo-random entropy variation
    }
    return FALSE;
}
```

To track permanent registers across soft-resets without data sector overflows or save corruption, the tracking fields are explicitly written into the permanent structural definition of SaveBlock2 in include/global.h:

```
struct SaveBlock2
{
    // ... vanilla fields preserve boundary limits ...
    u32 shinyShardsCount;       // Persistent global shard tracking architecture
    u16 currentChainSpecies;    // Tracks current species ID index targeted
    u8  currentChainCount;      // Bound tracker to max 255 value capacity
    u8  breedingPityCounter;    // Caches dry hatching counters for soft-ramping
};
```

4. Curated Regional Pokédex Allocation & Utility Hierarchy

To guarantee full stability of the overworld follower engine and eliminate save-block footprint bloat, the launch version caps total species availability to a balanced, custom regional pool of **400 to 500 total entries**. The pool breaks down into a distinct architectural split: 60% native Hoenn allocation (guaranteeing immediate, flawless custom walking sprite compatibility), 20% specialized cross-generational evolutionary expansions (e.g., Kingambit, Annihilape), and 20% exotic endgame reward suites.

Because tactical coordination is required to capture dangerous targets, specific utility anchors are locked into the early-game dex hierarchy:

Smeargle	Ditto Farming / Precision Capture Integration	False Swipe, Spore, Thief, Substitute
Breloom	Standard High-Velocity Physical Damage Catcher	Spore, False Swipe, Substitute, Seed Bomb
Spinda	Alpha Neutralization / Boss Suppression Engine	Skill Swap (Contrary Variant), Focus Sash Utility

5. Two-Phase Starter Shunting UI Architecture & Mode Gates

Phase 1: Fresh Save File Parameters (The Purist Track)

Phase 2: New Game Plus Framework (The Shunting Sandbox)

6. Expanded DexNav Interface & Chaining Controls

- **Top Right Component (Search Level):** Increments permanently by 1 per species sight line, providing a persistent baseline shiny probability adjustment over time.

- **Center Interface (Attribute Analysis):** Dynamically reads hidden attributes of rustling overworld nodes, displaying guaranteed 31 IV Potential Stars, Hidden Abilities, and rare Egg Move lines in illuminated gold typography.
- **Bottom Right Component (Roll-Budget Readout):** Outputs the active math formula dynamically in real-time as a visual fraction (e.g., Base 1x + Chain 3x = 4 Rolls), granting immediate systemic transparency.

Strict Operational Chaining Logic

7. Horde Transactional Blocks & Poké Ball Safety Locks

8. New Game Plus Variable Migration Contracts

- **Permanent Chain Lock:** In NG+, the standard automatic zero-reset parameter upon hitting a 100-count chain is deactivated. The DexNav permanently clamps and 'locks' at an optimal 99-chain tier, allowing continuous hunting at the hard 1/512 ceiling indefinitely until broken by a standard traversal violation.
- **Currency Retention:** Caches a proportional 25% allocation of total accumulated Shiny Shards directly into the starting wallet.

- **Global Dex Preservation:** Global Encounter Level registers remain completely intact, preserving baseline shiny adjustments across the transition.

End of System Specification Blueprint
This document serves as an immutable engineering framework. Modifications to core logic parameters must be approved via project architecture reviews.